LAB 2: FREQUENCY DOMAIN AND TRANSFER FUNCTIONS

```
%---------------------------------------
% Fill in your student Name and ID.
Students.Name = '';
Students.ID = '';
%---------------------------------------
```

0. Objectives

The purpose of this lab is to overview the basic **MATLAB** functions in linear systems and control. At the end of this lab, you will be able to plot the step and impulse response of a rational transfer function and analyze with first and second order transfer functions.

1. Introduction

The first step in the design of a control system is to identify a mathematical model of the system. Such a mathematical model could be derived either from physical laws or from experimental data. For Linear Time Invariant (LTI) systems, we only need to identify the impulse response or it's frequency-domain representation: the transfer function.

In this lab, we introduce how to model transfer functions in **MATLAB**. There are various methods to define a transfer function in **MATLAB**. As an example, suppose we have a system described by the following constant coefficient linear differential equation:

$$a_2 \frac{d^2y(t)}{dt} + a_1 \frac{dy(t)}{dt} + a_0 y(t) = b_1 \frac{du(t)}{dt} + b_0 u(t)$$

We know that the transfer function of this system is

$$H(s) = \frac{Y(s)}{U(s)} = \frac{b_1 s + b_0}{a_2 s^2 + a_1 s + a_0}$$

Suppose $a_0 = 2$, $a_1 = 3$, $a_2 = 1$, $a_3 = 1$, $b_0 = 2$, $b_1 = 2$.

The first method of specifying the function is to literally type out this expression. We first define the symbol s using the **tf** function and then define the transfer function H in terms of s. For example:

```
% Assign values to the coefficients
a2 = 1;
a1 = 3;
a0 = 2;
b1 = 2;
b0 = 6;
% Define s symbolically in frequency domain
s = tf('s');
```

Then we can directly define the transfer function as follows.

```
H1 = (b1 * s + b0) / (a2 * s^2 + a1 * s + a0)
```

```
H1 =


     2 s + 6
  -------------
  s^2 + 3 s + 2


Continuous-time transfer function.
```

The second method is to directly specify the polynomials of the numerator and the denominators.

```
% Define the numerator
num = [b1, b0];
% Define the denominator
den = [a2, a1, a0];
% Define the transfer function
H2 = tf(num, den)
```

```
H2 =


     2 s + 6
  -------------
  s^2 + 3 s + 2


Continuous-time transfer function.
```

Another way to define the transfer function is to use the gain, poles and zeros. Let's reconsider the above transfer function in pole-zero form.

$$H(s) = 2\frac{s+2}{(s+1)(s+2)}$$

We can specify the transfer function in the pole zero form using the following code.

```
k = 2; % The gain
z = [-3]; % The set of zeros
p = [-1, -2]; % The set of poles
H3 = zpk(z, p, k) % Define the transfer function
```

```
H3 =


    2 (s+3)
  -----------
```

```
  (s+1) (s+2)


Continuous-time zero/pole/gain model.
```

There are other forms of using the **zpk** function. Use **doc zpk** for more information.

It is possible to identify the poles and zeros of a **rational** transfer function using **tf2zp**:

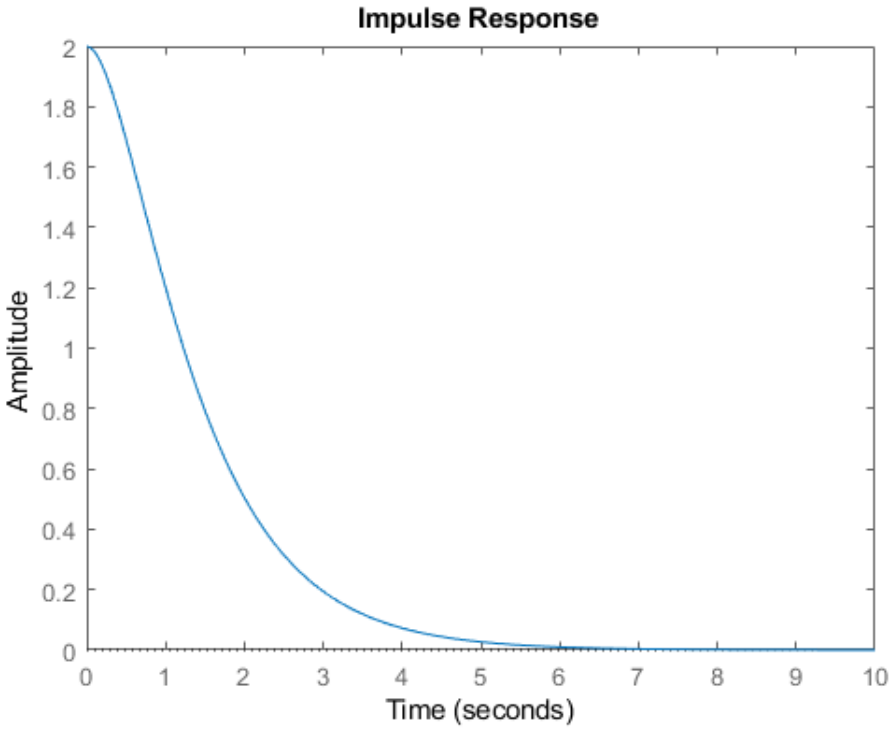
```
[z, p, k] = tf2zp(num, den)
```

```
z =
    -3
p =
    -2
    -1
k =
     2
```

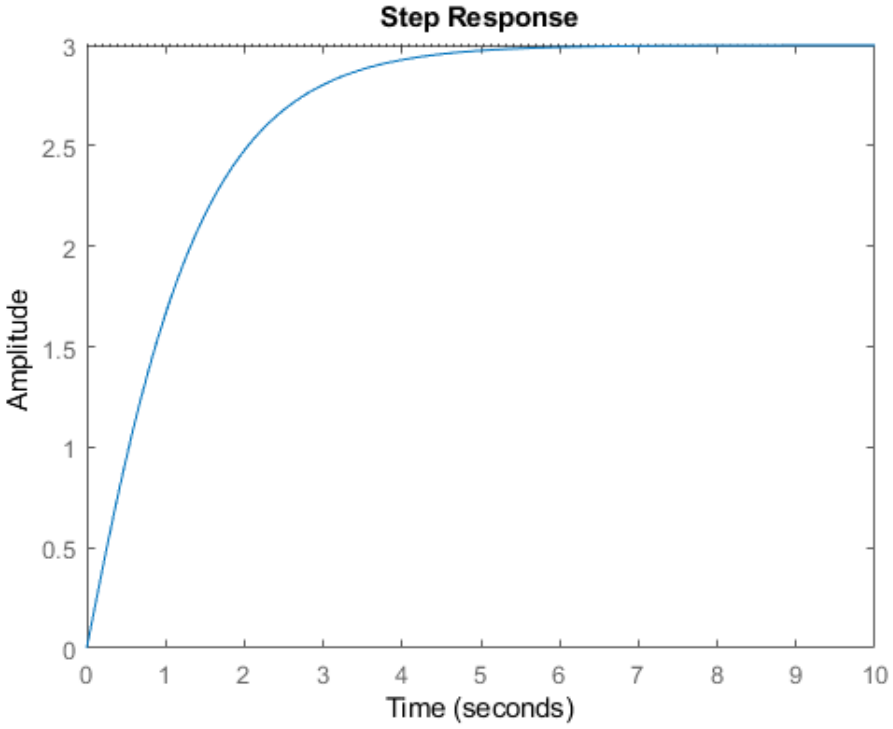
2. Plotting Impulse and Step Responses

To plot the impulse response of a transfer function, one can use **impulse(sys)**. To plot the step response, one can use **step(sys)**. As an example, we plot the impulse and step responses of the system defined in the previous section:

```
T = 10; % Final time value
figure;
impulse(H1, T)
```




```
figure;
step(H1, T)
```



To find the poles and zeros of a transfer function, use **`pole(system)`** and **`zero(systems)`**. To plot the pole-zero plot, use **`pzplot(system)`**. For example, for the system defined above, the poles are

```
pole(H1)
```

```
ans =

    -2

    -1
```

and the zeros are

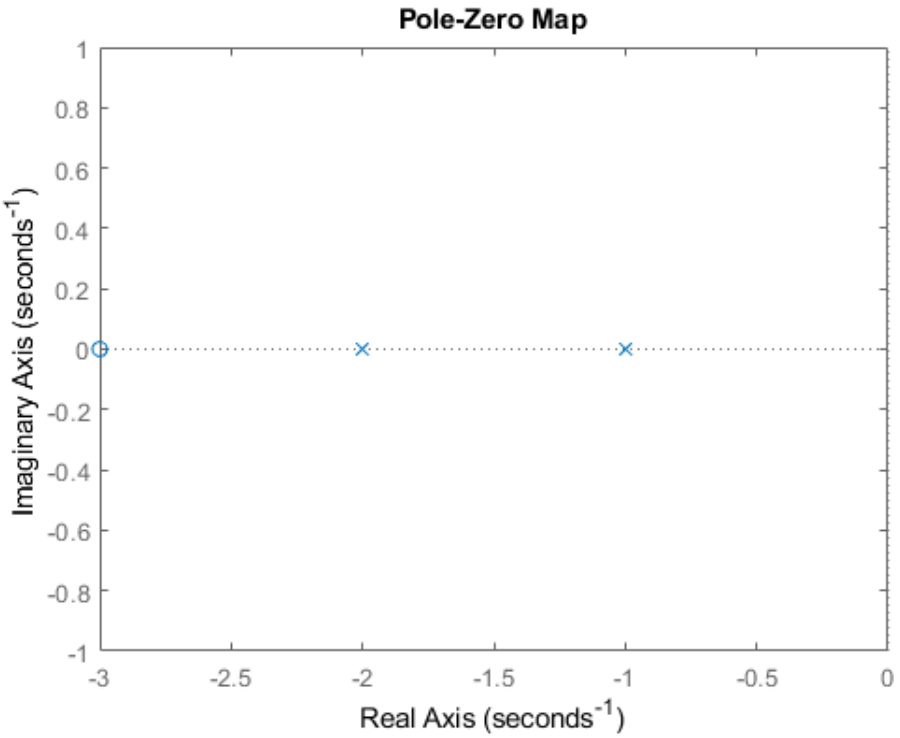
```
zero(H1)
```

```
ans =

    -3
```

We can show the pole-zero plot using

```
pzplot(H1)
```



Interconnecting systems

To get the transfer function for two systems connected in feedback, use

```
H = feedback(H1, H2) % Feedback or closed-loop interconnection of systems
```

```
H =

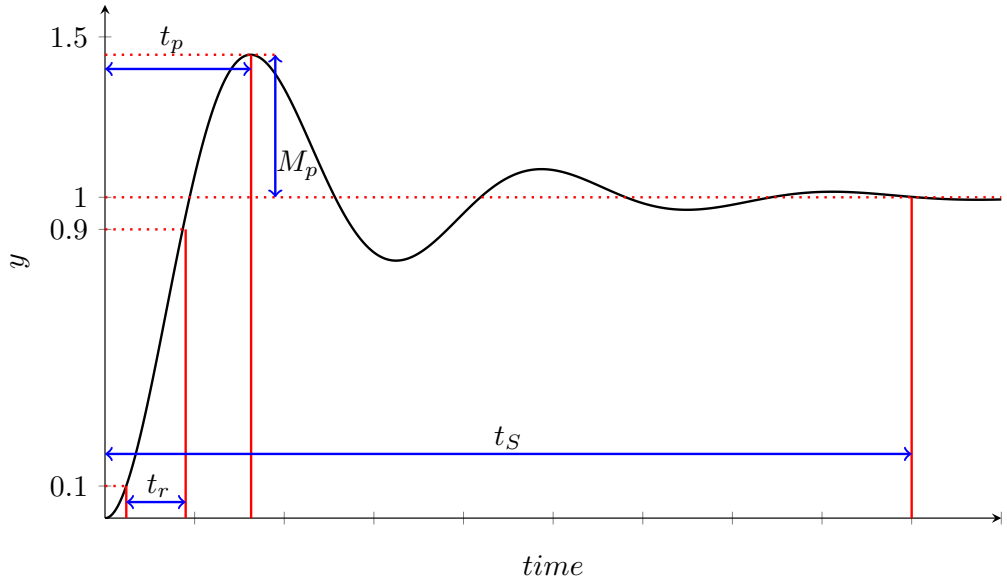
     2 s^3 + 12 s^2 + 22 s + 12
  --------------------------------
  s^4 + 6 s^3 + 17 s^2 + 36 s + 40

Continuous-time transfer function.
```

Use **`doc feedback`** for more information.

3. System Response characteristics

Consider the following step response of an unknown transfer function.



Depending on the location of the zeros and poles in a transfer function, it is possible to define a system response by the following characteristics:

- Rise time (t_r): is the time it takes the system to reach to rise from 10% to 90% of its final value.
- Peak time (t_p): is the time until the response hits its maximum overshoot.
- Overshoot (M_p): is the peak value of the step response of the system.
- Settling time (t_s): is the time for the response to settle within $\pm\Delta$ % of its final value.

It is possible to find these values in **MATLAB** given a transfer function. **stepinfo(sys)** computes the step-response characteristics for a dynamic system model. For example

```
stepinfo(H1)
```

```
ans =
  struct with fields:
        RiseTime: 2.4200
    SettlingTime: 4.1960
     SettlingMin: 2.7023
     SettlingMax: 2.9974
       Overshoot: 0
      Undershoot: 0
            Peak: 2.9974
        PeakTime: 7.3222
```

Question 1 (2 marks)

For $\xi = \{0.25, 0.75, 1, 2\}$, plot the step response **(for 25 sec)** and identify the characteristics of the transient response of the following system

$$G(s) = \frac{1}{s^2 + 2\xi s + 1}$$

```
% For xi = 0.25
%
```

```
% For xi = 0.75
%
```

```
% For xi = 1
%
```

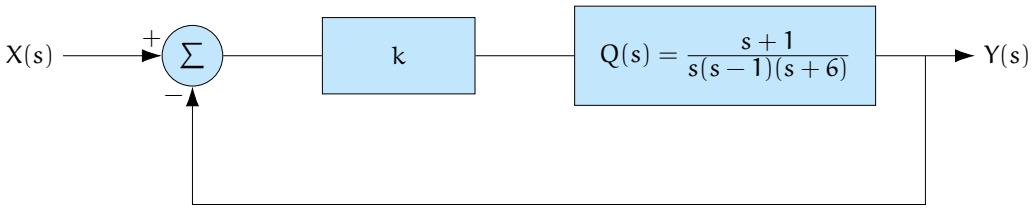
```
% For xi = 2
%
```

Question 2 (4 marks)

Consider the following open-loop unstable system.

$$Q(s) = \frac{s+1}{s(s-1)(s+6)}$$

A constant gain k (controller) is used to stabilize this system. For different values of $k \in \{10, 20, 50, 100\}$, plot the step response **(for 15 seconds)** of the closed-loop system and identify its characteristics using **stepinfo**.




```
% Define the transfer function
%
```

```
% k = 10
%
```

```
% k = 20
%
```

```
% k = 50
%
```

```
% k = 100
%
```

Question 3 (4 marks)

Consider the closed-loop transfer function in Question 2.

$$T(s) = \frac{kQ(s)}{1 + kQ(s)}$$

We would like to approximate this system to a second order system of the form:

$$G(s) = \frac{\frac{k}{|p_3|}(s - z_1)}{(s - p_1)(s - p_2)}$$

1. For $k = \{10, 20\}$, find p_1, p_2 and p_3, where p_1 and p_2 are the dominant poles (i.e., poles with the smallest absolute real values) and p_3 is the ineffective pole(i.e., the pole with the largest absolute real value).
2. For $k = \{10, 20\}$, plot the step response of the new second order system for **15 sec** and identify its characteristics using **`stepinfo`**.
3. For $k = \{10, 20\}$, compare the step response with those obtained in Question 2. What do you observe?

   - 1)

```
% Find the pole(s) and zero(s) of the two transfer functions
%



%
```

Transfer functions with dominant poles

```
% k = 10
%
```

```
% k = 20
%
```

- 2)

```
% k = 10
%
```

```
% k = 20
%
```

- 3)

```
% k = 10
%
```

```
% k = 20
%
```

```
% [Answer here]
```